

Review

Inference for a Mean (One-Sample t -Interval)

For an i.i.d. sample $X_1, \dots, X_n$ from a population X with unknown mean $\mathbb{E}[X] = \mu$ and variance $\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sigma^2$, we want to infer μ .

We will use the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

as a natural and intuitive estimator. Note that an **estimator** is a function of the random variables in the sample. For a particular observed sample, plugging in the observed values yields an **estimate**.

Note that the sample mean provides only a point estimate of μ , without accounting for estimation uncertainty. Since we are using a single random sample of finite size n , some deviation from the true value μ is expected. To quantify this uncertainty, we construct an interval estimate, most commonly a confidence interval.

To do so, we recall the **Central Limit Theorem** (CLT), which states that the sampling distribution of the sample mean is approximately normal when the sample size is sufficiently large. The power of the CLT is its broad applicability—it holds under fairly general conditions on the population distribution of X . Specifically, we have

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right),$$

that is, the sample mean estimator is centered at the true value μ , and its deviation approximately follows a normal distribution with variance $\frac{\sigma^2}{n}$. Therefore, the larger the sample size n , the smaller the variability of $\bar{X}_n$, which makes sense. One question remains: it is unlikely that we do not know μ but do know σ^2 . To address this, we again estimate it using the sample variance.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

To compensate for the fact that we do not know σ^2 and instead estimate it using s^2 , we inflate the estimation precision and obtain the following distributional result:

$$\frac{\bar{X}_n - \mu}{\frac{s}{\sqrt{n}}} \approx t_{df=n-1},$$

which is the Student- t distribution with $n - 1$ degrees of freedom (we lose one degree of freedom due to estimating the mean). As a result, a $(1 - \alpha)100\%$ confidence interval for μ is

$$\bar{X} \pm t_{n-1, 1-\alpha/2} \cdot \frac{s}{\sqrt{n}},$$

where, by abuse of notation, $t_{n-1, 1-\alpha/2}$ denotes the $(1 - \alpha/2)$ quantile of the Student- t distribution with $n - 1$ degrees of freedom.

Inference for Group Means (ANOVA)

For a sample

$$X_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n_i,$$

where

$$X_{ij} = \mu_i + \varepsilon_{ij},$$

and the ε_{ij} are i.i.d., we want to compare the group means $\mu_1, \mu_2, \dots, \mu_k$.

Here we assume that all groups share the same variance:

$$\sigma^2 \quad \text{is constant across groups.}$$

We estimate σ^2 using the **Mean Squared Error** (MSE):

$$\hat{\sigma}^2 = MSE = \frac{SS_W}{N - k},$$

where $SS_W = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$, $N = \sum_{i=1}^k n_i$ is the total sample size, and k is the number of groups.

For group i , the sample mean is $\bar{X}_i$ with group size n_i . The standard error (SE) of $\bar{X}_i$ under ANOVA is

$$SE(\bar{X}_i) = \sqrt{\frac{MSE}{n_i}}.$$

Thus, a $(1 - \alpha)100\%$ confidence interval for the group mean μ_i is

$$\bar{X}_i \pm t_{N-k, 1-\alpha/2} \cdot \sqrt{\frac{MSE}{n_i}}.$$

Inference for Regression Function (Simple Linear Regression)

Simple Linear Regression Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim \text{i.i.d. } N(0, \sigma^2)$$

- Estimators for the regression slope and intercept:

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

- Confidence interval for slope β_1 :

$$\hat{\beta}_1 \pm t_{n-2, 1-\alpha/2} \cdot \text{SE}(\hat{\beta}_1), \quad \text{SE}(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (X_i - \bar{X})^2}}$$

- Confidence interval for mean response at X_0 :

$$\hat{Y}_0 \pm t_{n-2, 1-\alpha/2} \cdot s \sqrt{\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum (X_i - \bar{X})^2}}$$

- Prediction interval for new observation at X_0 :

$$\hat{Y}_0 \pm t_{n-2, 1-\alpha/2} \cdot s \sqrt{1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum (X_i - \bar{X})^2}}$$

From Classical Inference to Dependent Data

Classical methods reviewed above all assume independent errors. In time series, it is often the case that data exhibit temporal dependence, as we will see in concrete examples later. Such dependence inflates or changes uncertainty; the effective sample size can be much smaller than n . We must *model* temporal dependence to obtain valid inference and accurate forecasts.

An Overview of Time Series Analysis

In this section, we begin our study of **time series analysis**. Rather than starting immediately with models, we first examine several real-world time series to understand the types of patterns and questions that arise when data are collected over time.

We will then develop the basic framework for time series data:

$$\text{examples} \longrightarrow \text{data structure} \longrightarrow \text{models} \longrightarrow \text{objectives}.$$

In particular, we will:

- examine several **examples of time series data** and identify common temporal patterns;
- introduce the general structure

$$\{X_t : t \in \mathcal{T}\},$$

and distinguish between **discrete-time** and **continuous-time** processes;

- introduce the basic idea of a **time series model** and discuss how it differs from more familiar statistical models, particularly through dependence across time;
- discuss common objectives of time series analysis, including **describing temporal patterns, understanding dependence, modeling dynamics, and forecasting future observations**.

The central idea is that in time series data, **time ordering is part of the data structure**: observations close in time are often related, and this dependence contains useful information rather than simply being a nuisance.

Time Series Data Examples

We now move from **independent observations** to **time series data**:

$$X_1, X_2, \dots, X_T,$$

where observations are recorded **sequentially over time**. Unlike independent data, the **ordering matters**, and observations at different time points may be dependent.

- Lake Huron Time Series (Brockwell and Davis, 1991)

Example: Lake Huron Water Levels.

The built-in R dataset `LakeHuron` contains annual measurements of the **level of Lake Huron (in feet)** from 1875 to 1972.

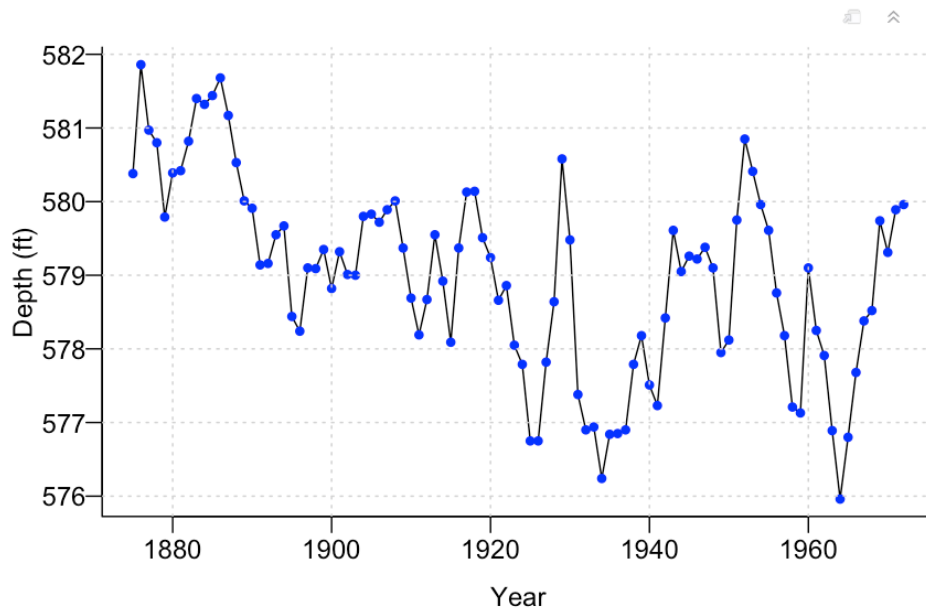
$$X_t = \text{Lake Huron water level in year } t.$$

We begin by loading the data and examining its time-series structure.

```

{r}
par(mar = c(3.2, 3.2, 0.5, 0.5), mgp = c(2, 0.5, 0), bty = "L")
data(LakeHuron)
plot(LakeHuron, ylab = "Depth (ft)", xlab = "Year", las = 1)
points(LakeHuron, cex = 0.8, col = "blue", pch = 16)
grid()

```



- CO₂ Concentration

The built-in R dataset `co2` contains **monthly atmospheric CO₂ concentrations**, measured in parts per million (ppm).

Unlike the annual `LakeHuron` series, the observations are recorded **monthly**. Thus, we can examine variation at two time scales:

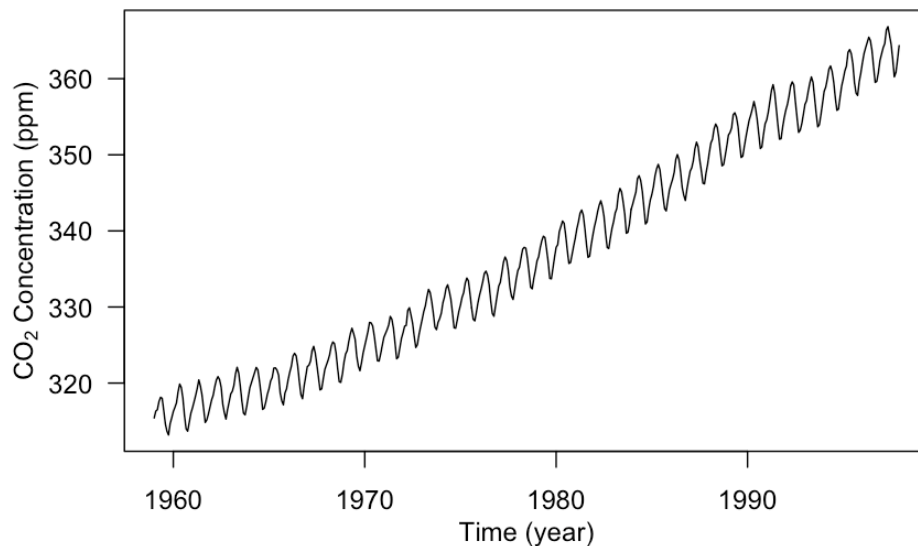
across years and within each year.

This higher sampling frequency allows us to see not only long-term changes, but also systematic patterns that repeat within each year.

```

{r}
data(co2)
par(mar = c(3.8, 4, 0.8, 0.6))
plot(co2, las = 1, xlab = "", ylab = "")
mtext("Time (year)", side = 1, line = 2)
mtext(expression(paste("CO"[2], " Concentration (ppm)")), side = 2, line = 2.5)

```



- Apple Daily Log Returns

Financial data provide another important type of time series.

- Stock prices: P_t
- Often analyze **returns** rather than prices
- Returns measure **relative changes** in price over time

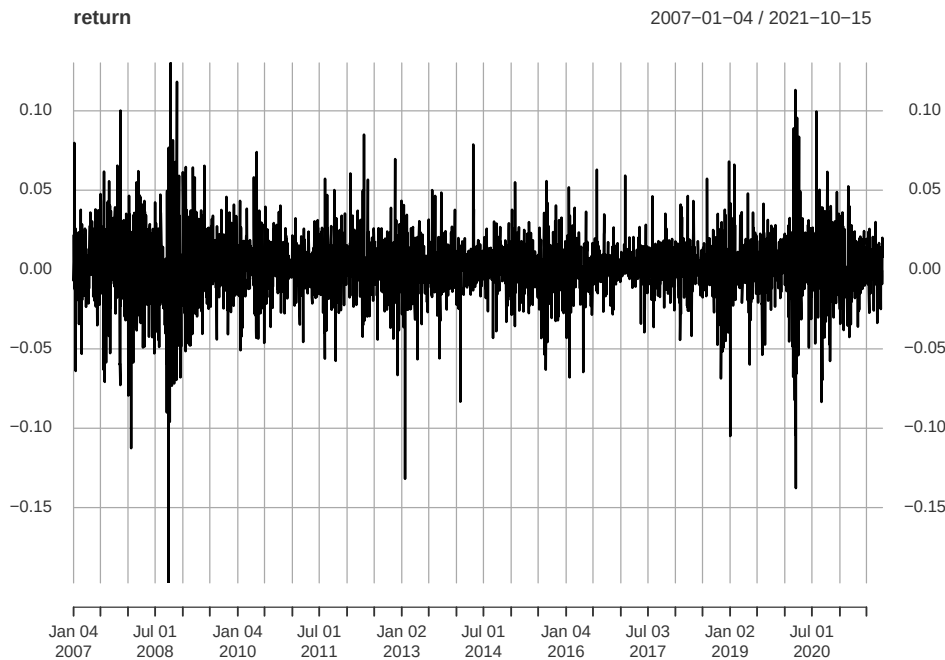
For a closing price P_t , define the **daily log return** as

$$r_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right).$$

R example: Apple (AAPL)

daily prices $\longrightarrow$ log returns.

We use the **quantmod** package to obtain daily AAPL stock prices from Yahoo Finance and compute the corresponding log returns.



- Global Mean Land Temperature Anomalies

So far:

one variable observed over time $\longrightarrow$ **univariate time series**.

In many applications, we observe **several related variables** over the same time period:

$$\mathbf{X}_t = \begin{pmatrix} X_{1t} \\ X_{2t} \\ \vdots \\ X_{pt} \end{pmatrix}, \quad t = 1, \dots, T.$$

This gives a **multivariate time series**.

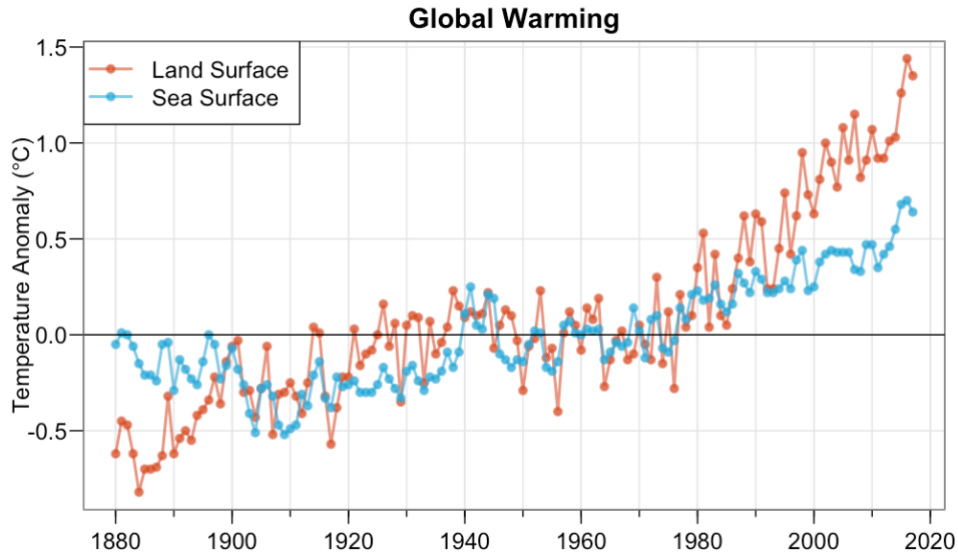
R example: Global temperature anomalies

The `astsa` package contains annual temperature anomaly series for

- **land surface temperature**, and
- **sea surface temperature**.

We plot the two series together to examine

individual temporal patterns + how the two series evolve jointly over time.

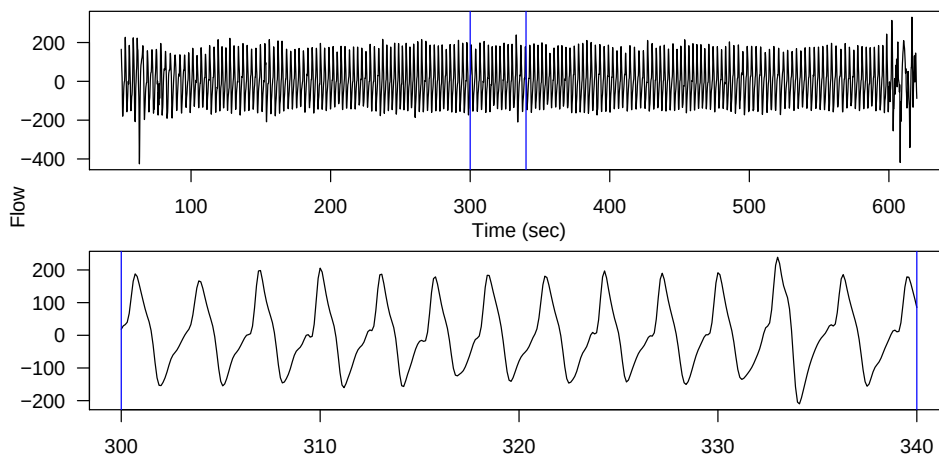


- Sleep Airflow Signal

The sleep airflow signal also exhibits a clear **oscillatory pattern**, reflecting the repeated inhalation–exhalation breathing cycle. Compared with the previous CO_2 example, however, the oscillations occur at a much **higher frequency**. When the signal is displayed over a relatively long time interval, the individual cycles are compressed together, so we often need to **zoom in** to examine the local oscillatory structure more carefully.

The pattern is also more complicated than the relatively regular oscillations seen in the CO_2 series. The **amplitude**, **waveform**, and **spacing between cycles** can all vary over time. Thus, the signal is oscillatory, but it may not be well represented by a single fixed frequency or a simple sinusoidal wave.

This type of behavior motivates **time–frequency analysis**. Rather than only asking which frequencies are present in the signal, we may also want to study *how the frequency content changes over time*. This provides a way to characterize local changes in breathing patterns that may not be apparent from the raw time-series plot alone.



- Simulated Time Series

Our final example is a **simulated stochastic process**, rather than an observed dataset. Consider the **random walk**

$$X_t = X_{t-1} + W_t, \quad W_t \stackrel{\text{i.i.d.}}{\sim} N(0, 1),$$

with, for example, $X_0 = 0$.

Equivalently,

$$X_t = \sum_{j=1}^t W_j.$$

Key idea:

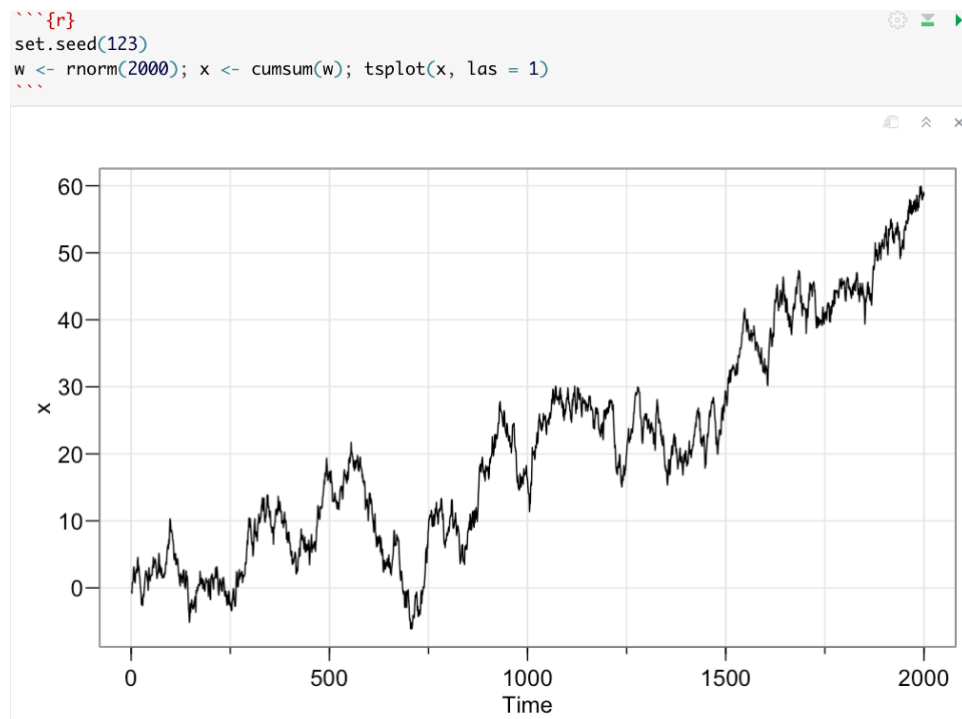
current value = previous value + new random shock.

Thus, each new observation is obtained by adding a **random increment** W_t to the previous value X_{t-1} .

This simple model illustrates an important feature of time series:

X_t depends directly on X_{t-1} .

Hence, observations over time are **not independent**, even though the increments W_t are independent.



What is a Time Series?

A time series is a collection of observations $\{y_t : t \in T\}$ taken sequentially in time t , where the index set T denotes the time points. These observations can be gathered at fixed (equidistant) time points (e.g., $T = \{0, 1, 2, \dots\}$), at irregular time points, or even over an entire time interval (e.g., $T = [0, T]$).

Different types of time sampling require different approaches to data analysis. In this course, we will focus on some of the most commonly used methods for dealing with the first scenario: **discrete-time** equal-spaced time series. A discrete-time time series might be intrinsically discrete (e.g., number of planes departing ATL every day) or might arise from an underlying **continuous-time** time series via

- **Sampling:** For example, instantaneous wind speed
- **Aggregation:** Such as daily accumulated precipitation amount
- **Extrema:** Like the daily maximum temperature

It is worth noting that the methods for continuous-time time series (where T represents an entire interval) are useful in situations where data were observed at irregular times.

Depending on the type of value that y_t can take, we have:

- **Real-valued:** a value in $\mathbb{R}$ or a subset thereof (e.g., temperature)
- **Complex-valued** (some applications in electrical engineering)
- **Non-negative integer** (See Jia et al. (2021) for a recent development for modeling such kind of count time series data)
- **Categorical** (e.g., outcome of scheduled basketball match: win, lose, cancellation)
- **Circular** (e.g., wind direction. See Breckling (2012))

Exploratory Time Series Analysis

Suppose we have time series data $\{y_t : t \in T\}$. We will start with a **time series plot** of y_t against t . It is important to label the axes accurately (i.e., the time unit on the x-axis, variable name and its unit on the y-axis) and adjust the aspect ratio to create an aesthetically pleasing time series plot. In this exploratory data analysis (EDA) stage, we look the following:

1. Are there abrupt changes? (e.g., shifts in mean and/or variance)
2. Are there **outliers**? (i.e., values that are unusual relative to the rest of the data)
3. Is there a need to transform the data (e.g., should we apply a transformation to stabilize the variance)?
4. Features of the time series: trend, seasonal components, and noise process

Decomposition: Trend, Seasonality, Noise

- Trends (μ_t)

One can think of trend, μ_t , as representing continuous changes, usually in the mean, over longer time scales. *The essential idea of a trend is that it should be smooth.* Typically, the form of the trend is unknown and needs to be specified and then estimated for the data. When the trend is estimated and then removed, we obtain a “detrended” series

- Seasonal components (s_t)

A seasonal component, s_t , constantly repeats itself in time, (i.e., $s_t = s_{t+kd}$ for all t and k). Sometimes the seasonality exhibits a very clear structure form (e.g., a sin or cosine wave), while at other times, the structure is more complex (see the sleep airflow data example used in Huang et al. (2022)).

- Noises (η_t)

The noise process, η_t , is the component that is neither trend nor seasonality. Our focus will be on identifying plausible (typically stationary) statistical models for this process.

There are two common approaches to combine these components:

$$\begin{aligned}\text{Additive: } y_t &= \mu_t + s_t + \eta_t, \\ \text{Multiplicative: } y_t &= \mu_t s_t \eta_t.\end{aligned}$$

If $y_t > 0$, then $\log y_t = \log \mu_t + \log s_t + \log \eta_t$ converts multiplicative to additive on the log scale.

Time Series Models

A time series model is a probability model for $\{Y_t : t \in T\}$. Unlike i.i.d. settings with many independent samples, we typically observe only a *single* T -dimensional realization (one sample path). Thus inference is a one-sample, high-dimensional problem rather than a many-sample, low-dimensional one.

To make inference feasible, we impose temporal structure—most commonly stationarity (and often ergodicity)—so that time shifts create effective “replications” across lags. For example, the pairs $\{(Y_t, Y_{t+h})\}_t$ provide repeated information about dependence at lag h , enabling estimation via sample autocovariances.

We therefore aim for a simple *stationary* model for the noise process η_t after removing nonstationarity (trend/seasonality):

- **Second-order stationarity.** $E[\eta_t] = 0$ for all t and $\text{Cov}(\eta_t, \eta_{t'}) = \gamma(t' - t)$ depends only on the lag.
- **Autocorrelation.** The autocorrelation function (ACF) $\rho(h) = \gamma(h)/\gamma(0)$ summarizes linear dependence by lag h .

Lag plots and ACF. Lag plots of (y_t, y_{t+h}) reveal association at short lags; the ACF formalizes this across many lags and guides model choice.

Objectives of Time Series Analysis

Modeling: Explain dependence structure (e.g., trend + autocorrelation) and support inference (e.g., trend significance).

Forecasting: Predict future values with uncertainty quantification.

Adjustment: Remove seasonality/holiday effects to clarify underlying movement.

Simulation: Generate synthetic paths from a fitted model to understand behavior or assess procedures.

Control: Adjust inputs/parameters so outputs meet a standard (e.g., SPC/QC).

Time Series Analysis in the Era of AI

Time series analysis has a long history in statistics, but it has also become an important area of modern machine learning (ML) and artificial intelligence (AI). The models and computational tools are changing rapidly, but the fundamental statistical questions remain similar.

At a high level, we observe a history

$$Y_1, Y_2, \dots, Y_T$$

and want to learn something about the underlying temporal process. In a forecasting problem, for example, we may want to predict

$$Y_{T+1}, Y_{T+2}, \dots, Y_{T+h},$$

where h is the **forecast horizon**.

Thus, a basic forecasting problem can be summarized as

past observations	→	learn temporal structure	→	future prediction.
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The central challenge is that observations collected over time are generally **not independent**. Information about the future is contained partly in the dependence between past and future observations.

From Classical Time Series Models to Modern AI

Classical time series analysis typically begins by explicitly specifying a probabilistic model for the temporal dependence.

Examples that we will encounter include:

- autoregressive and moving-average models (ARMA/ARIMA);
- exponential smoothing methods;
- state-space models;
- GARCH and related models for time-varying volatility.

For example, a simple first-order autoregressive model has the form

$$Y_t = \phi Y_{t-1} + \varepsilon_t,$$

where ε_t represents a new random innovation at time t .

The parameter ϕ has a direct interpretation: it describes how strongly the current observation is related to the previous observation.

Classical modeling philosophy:

specify temporal structure $\longrightarrow$ estimate parameters $\longrightarrow$ diagnose the fitted model $\longrightarrow$ forecast/infer.

These models are often relatively parsimonious and interpretable, and their probabilistic formulation provides a natural framework for **uncertainty quantification**.

Modern ML and AI approaches provide a different way of learning temporal structure. Examples include:

- tree-based and ensemble methods;
- recurrent neural networks (RNNs);
- long short-term memory networks (LSTMs);
- transformers;
- pretrained time-series foundation models.

Rather than requiring us to specify a simple dependence structure explicitly, these methods can learn complicated, nonlinear relationships from data.

Very roughly,

classical models: specify structure explicitly
--

whereas

modern ML/AI: learn flexible structure from data.

This distinction should not be interpreted too literally. Modern statistical models can also be highly flexible, and successful AI models often incorporate important structural assumptions. The difference is primarily one of **modeling philosophy, model flexibility, and scale**. In practice, there is substantial overlap between traditional statistical modeling and modern AI approaches.

Greater flexibility also comes with a cost. More flexible models can capture richer and more complicated patterns, but they typically require **more data** to estimate reliably. With limited data, highly flexible models may overfit and generalize poorly, while simpler models that incorporate appropriate structural assumptions may perform better. Thus, there is an important tradeoff among **model flexibility, data availability, and generalization**. More advanced or flexible models are not automatically better; the appropriate level of complexity depends on the available data and the goal of the analysis.

What Does AI Change?

Traditional time series models often emphasize:

- explicit assumptions about temporal dependence;
- relatively small numbers of parameters;
- statistical interpretability;
- probabilistic inference and uncertainty quantification.

Modern ML/AI models often emphasize:

- flexible nonlinear relationships;
- learning complicated temporal patterns automatically;
- potentially very large numbers of parameters;
- learning from large collections of time series.

One useful way to contrast the two approaches is

Traditional time series	$\iff$	Modern ML/AI
explicit structure	$\iff$	learned structure
parsimonious models	$\iff$	potentially large models
interpretability	$\iff$	flexibility
model-based UQ	$\iff$	data-driven prediction

However, this is **not a competition between “statistics” and “AI.”**

In practice, modern forecasting increasingly combines

statistical structure + machine learning + domain knowledge.

For example, we may use machine learning to represent a complicated nonlinear relationship while still using statistical ideas to construct prediction intervals, evaluate forecast errors, diagnose temporal dependence, or account for nonstationarity.

A New Direction: Pretrained Time-Series Models

A particularly important recent development is the use of **pretrained models** for time series. In traditional statistical modeling, we usually begin with a particular dataset:

target time series $\longrightarrow$ specify a model $\longrightarrow$ fit the model $\longrightarrow$ forecast.

For example, if we want to forecast Lake Huron water levels, we might fit an ARIMA or state-space model specifically to the Lake Huron observations.

Even when several related series are modeled simultaneously, as in multivariate or spatio-temporal analysis, the statistical model is generally constructed and fitted for the particular process under study.

The emerging **foundation-model paradigm** is different:

many time series $\longrightarrow$ pretrain a large model $\longrightarrow$ apply it to new time series.

The idea is that a sufficiently large and diverse collection of time series may contain temporal patterns that are useful across many applications. During pretraining, the model attempts to learn representations of these general temporal patterns.

Then, when presented with a new series,

$$Y_1, \dots, Y_T,$$

the pretrained model may use what it learned from other datasets to forecast

$$Y_{T+1}, \dots, Y_{T+h}.$$

In some cases this can be done without fitting a new model specifically to the target series. This is called **zero-shot forecasting**.

Important distinction.

Using many time series is *not* itself the new idea. Classical multivariate and spatio-temporal models have long analyzed multiple related time series jointly.

The important distinction is

pretraining + transfer to new time series.

That is, information learned from one large collection of time series is intended to be transferable to a new forecasting problem.

Why Classical Time Series Analysis Still Matters

The emergence of AI does not eliminate the fundamental problems of time series analysis. In fact, many of the statistical concepts developed in this course become even more important when evaluating complex forecasting models.

1. Temporal dependence

The fundamental question remains:

What does the past tell us about the future?

We need to understand how Y_t is related to

$$Y_{t-1}, Y_{t-2}, \dots$$

and how quickly that dependence weakens as the time lag increases.

Concepts such as autocovariance, autocorrelation, ARMA models, and state-space representations provide tools for describing this structure.

2. Uncertainty

A point forecast alone is usually not enough.

Suppose a model produces

$$\hat{Y}_{T+h}.$$

We also want to know how uncertain that prediction is.

A useful forecast therefore often takes the form

point forecast + prediction interval/distribution.

This distinction is especially important in applications where decisions depend not only on the expected future value but also on the possibility of extreme or unexpected outcomes.

3. Diagnostics

A model can produce reasonable predictions while still failing to capture important temporal structure.

After fitting a model, we therefore ask:

What structure has the model missed?

For classical models, we often examine residuals

$$e_t = Y_t - \hat{Y}_t$$

and ask whether they still contain temporal dependence, trend, seasonality, changing variance, or other systematic patterns.

The same principle applies to modern AI models: prediction accuracy alone does not tell us everything about whether the model has learned the relevant structure.

4. Validation

Time series cannot generally be validated by randomly shuffling observations into training and test sets.

Why?

Because

time ordering matters.

Using future observations to predict the past can create **information leakage** and produce unrealistically optimistic estimates of predictive performance.

A more natural evaluation strategy is

$$\underbrace{Y_1, \dots, Y_T}_{\text{training data}} \longrightarrow \underbrace{Y_{T+1}, \dots, Y_{T+h}}_{\text{future test data}}.$$

Thus, appropriate temporal validation is essential for both classical statistical models and modern AI forecasting systems.

5. Nonstationarity

Perhaps the most difficult issue is that the future may not behave exactly like the past.

We therefore need to ask:

What happens when the underlying process changes?

Changes may arise from trends, seasonality, structural breaks, changing variance, evolving relationships among variables, or changes in the data-generating mechanism itself.

A model—whether ARIMA, a transformer, or a large pretrained foundation model—can perform poorly when the future differs substantially from the data on which it was trained.

Take-Home Message

AI greatly expands the class of models that we can use for time series and allows information to be learned from increasingly large collections of data.

But the fundamental statistical questions remain:

Dependence: What does the past tell us about the future?

Uncertainty: How much can we trust a forecast?

Diagnostics: What structure has the model missed?

Validation: Will predictive performance generalize to the future?

Nonstationarity: What happens when the process changes?

Key message: AI changes the models we can use, but it does not remove the need to understand temporal dependence, quantify uncertainty, diagnose model misspecification, and validate predictions appropriately.

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